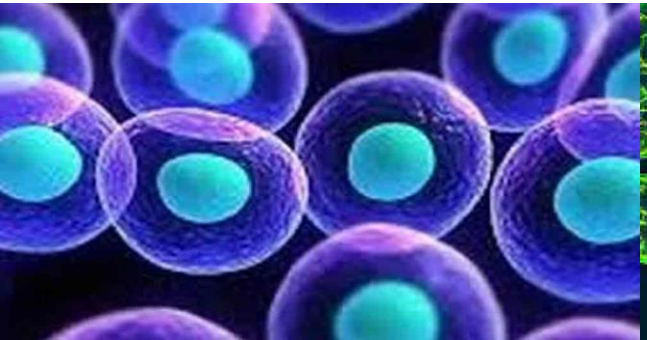
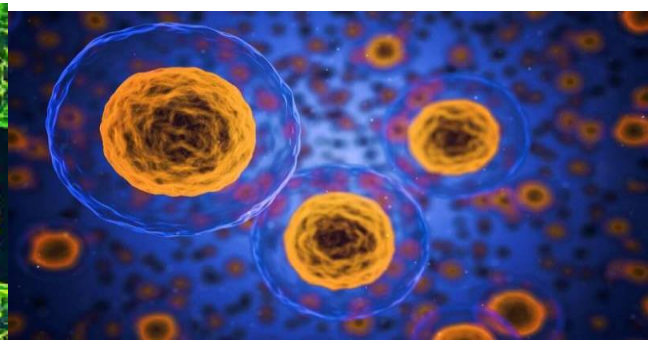
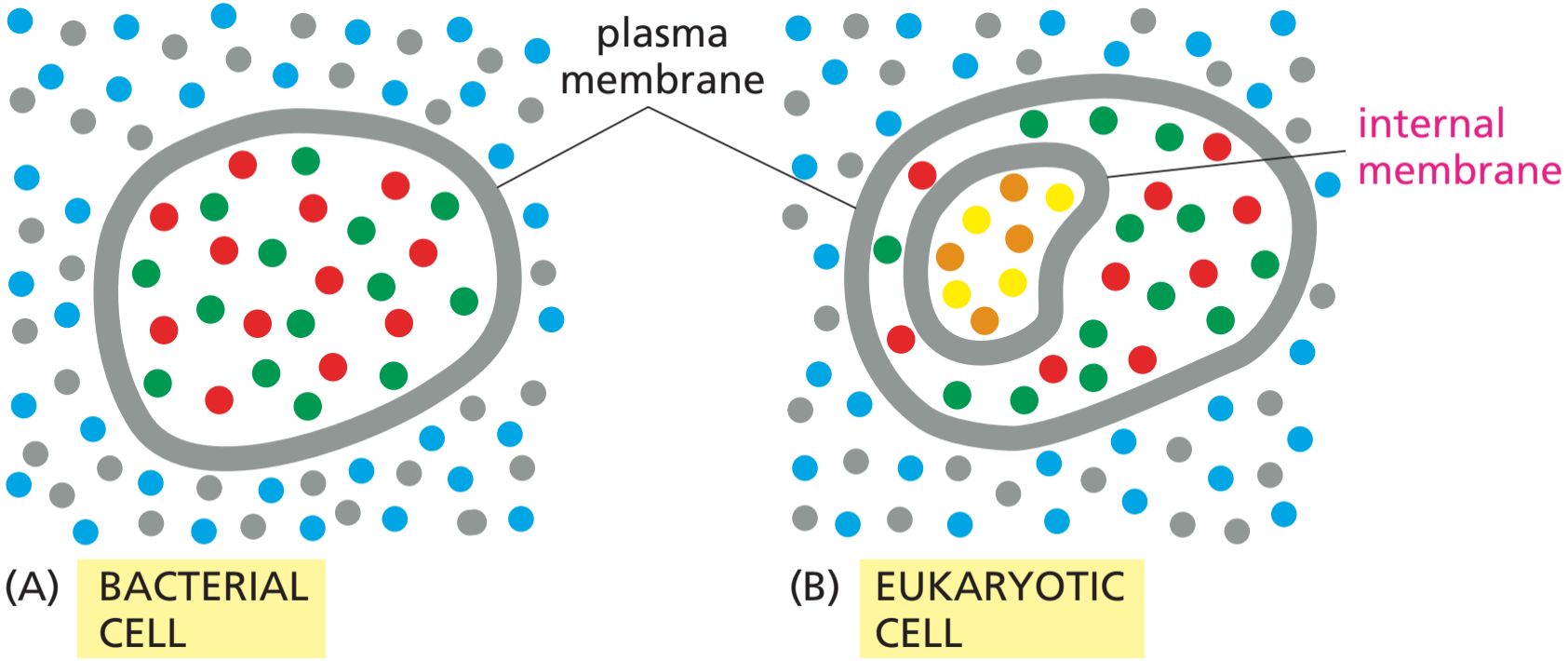
Part 4

Membranes and transport mechanism





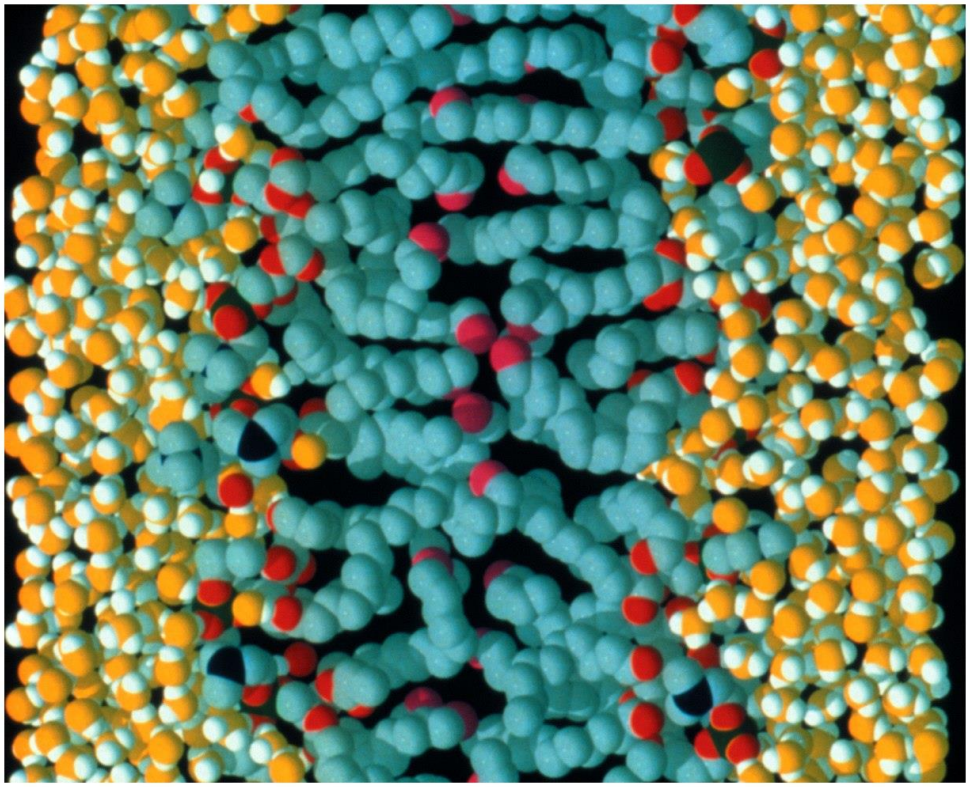


Cell membranes act as selective barriers.

The plasma membrane separates a cell from its surroundings, enabling its molecular composition to differ from its environment.

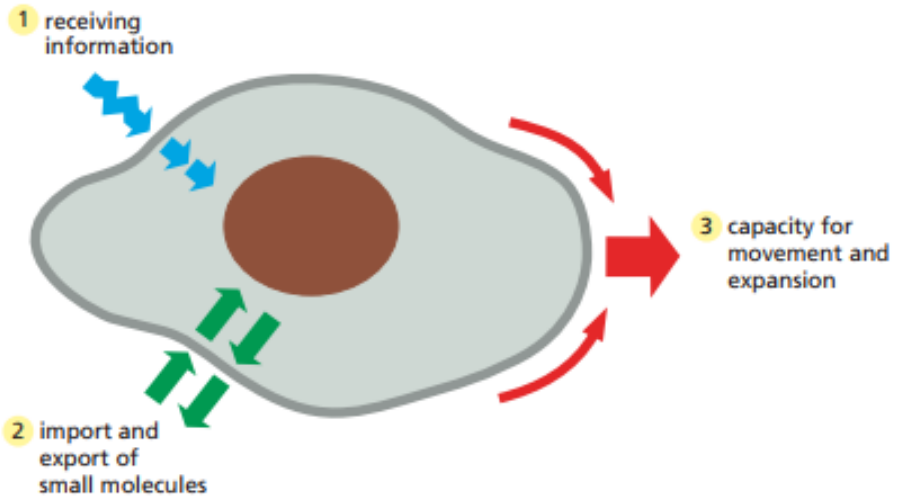
(a) In some bacteria, the plasma membrane is the only membrane.

(b) Eukaryotic cells also have internal membranes that enclose individual organelles. All cell membranes prevent molecules on one side from freely mixing with those on the other, as schematically indicated by the colored dots.



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BIOMEMBRANE STRUCTURE



The plasma membrane is involved in cell communication, import and export of molecules, and cell growth and motility.

(1) Receptor proteins in the plasma membrane enable the cell to receive signals from the environment;
(2) Transport proteins in the membrane enable the import and export of small molecules;
(3) The flexibility of the membrane and its capacity for expansion allow the cell to grow, change shape, and move.

The faces of cellular membrane

Internal cytosol (green stipple) and external environment (purple) define the cytosolic (red) and exoplasmic (black) faces of the bilayer.

Vesicles and some organelles have a single membrane and their **internal aqueous space** (purple) is topologically **equivalent to the outside** of the cell

Organelles enclosed by two membranes separated by a small intermembrane space.

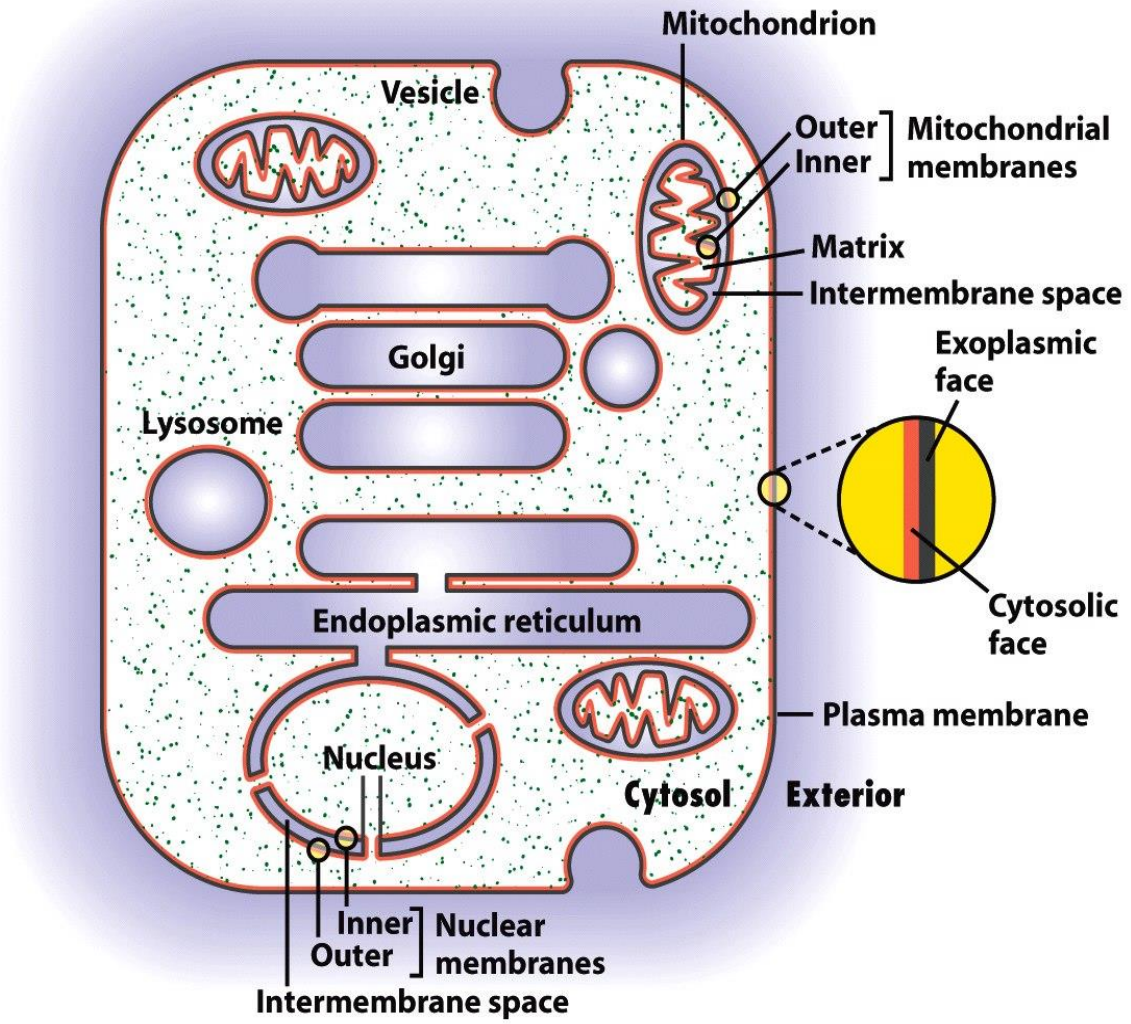


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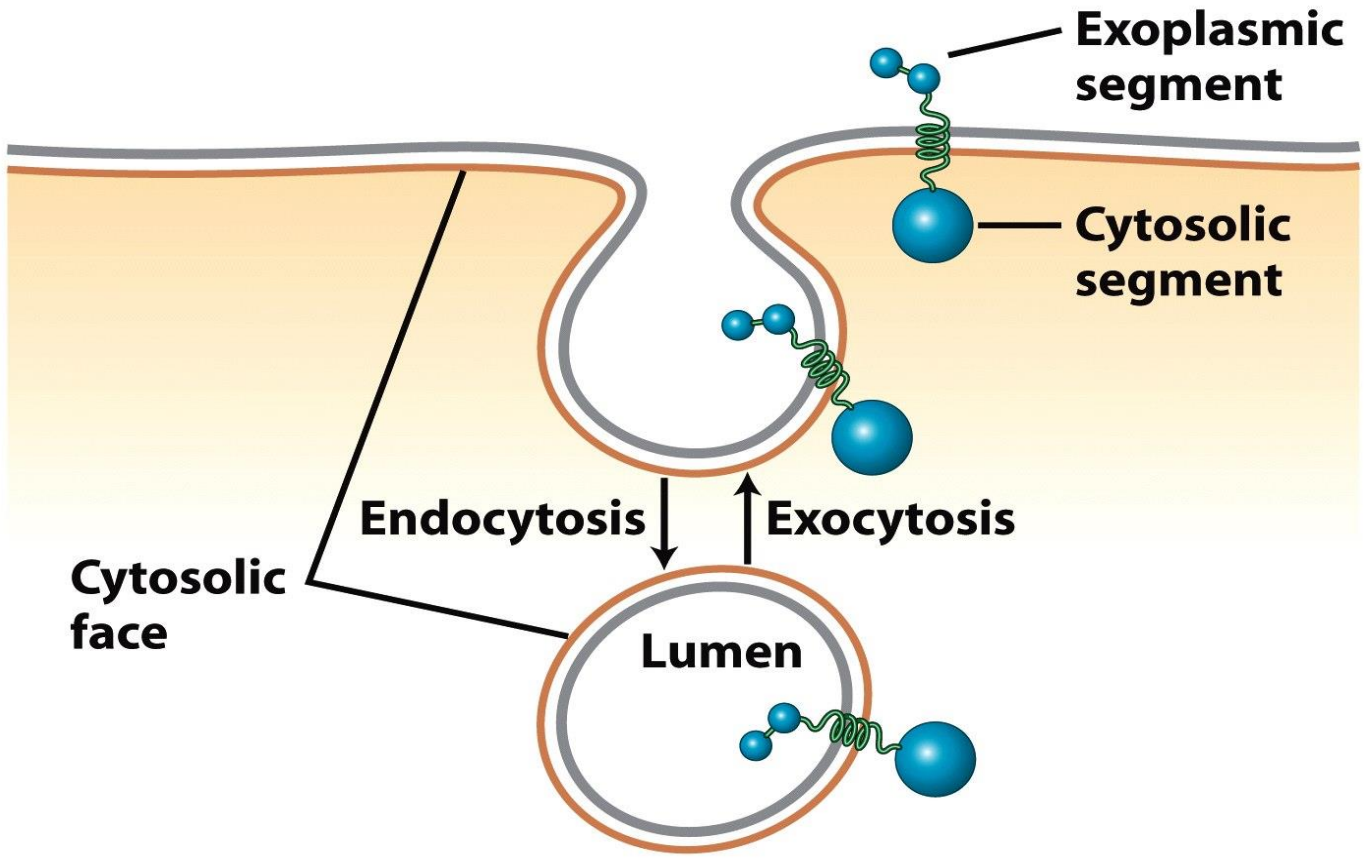


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Faces of cellular membranes are conserved during membrane budding and fusion

Structure

3 main components

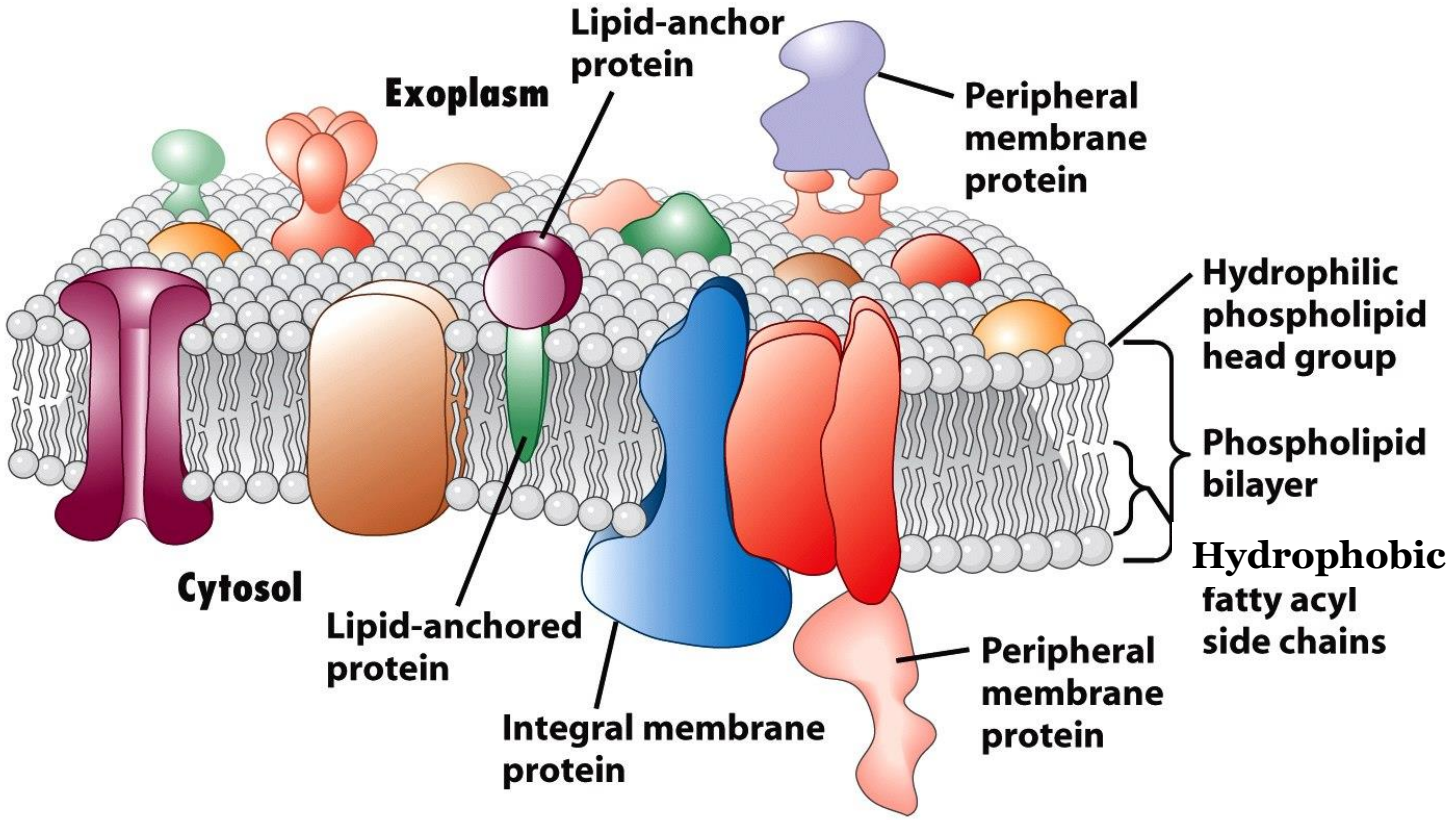


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Membrane lipids

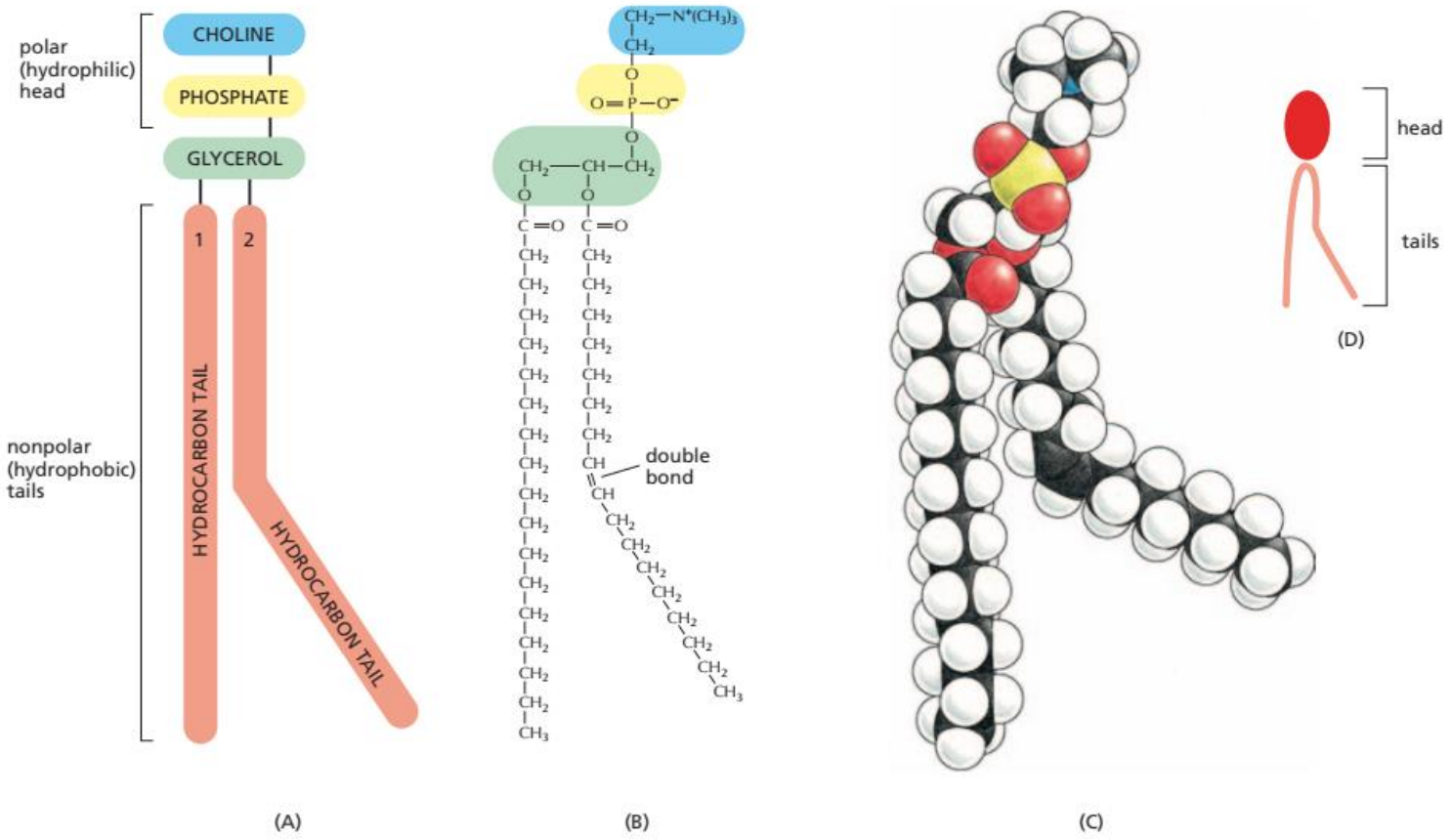


Figure 11–6 Phosphatidylcholine is the most common phospholipid in cell membranes. It is represented schematically in (A), as a chemical formula in (B), as a space-filling model in (C), and as a symbol in (D). This particular phospholipid is built from five parts: the hydrophilic head, which consists of *choline* linked to a *phosphate group*; two *hydrocarbon chains*, which form the hydrophobic tails; and a molecule of glycerol, which links the head to the tails. Each of the hydrophobic tails is a *fatty acid*—a hydrocarbon chain with a –COOH group at one end—which has been attached to glycerol via this group. A kink in one of the hydrocarbon chains occurs where there is a double bond between two carbon atoms. (The "phosphatidyl" part of the name of a phospholipid refers to the phosphate–glycerol–fatty acid portion of the molecule.)

Different types of membrane lipids

Phosphoglycerides are the principal building blocks of most biomembrane

Three classes of membrane lipids

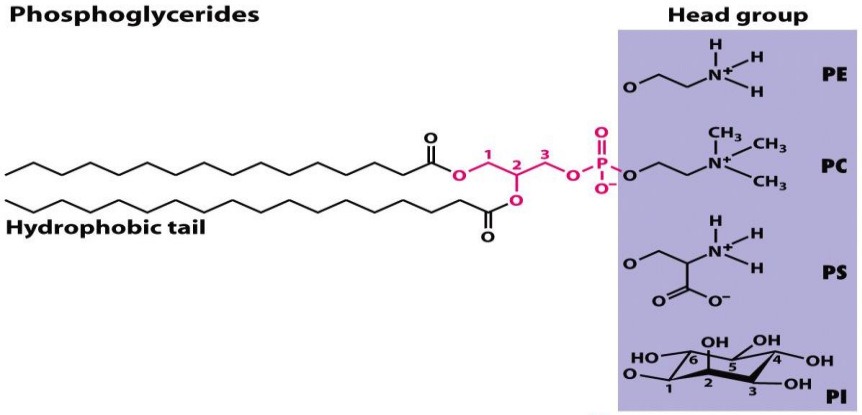


Figure 10-5a
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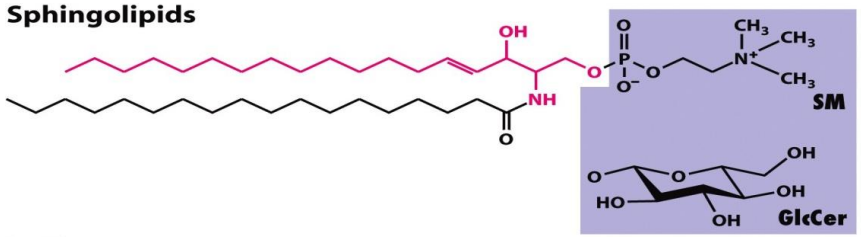


Figure 10-5b
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Most **phosphoglycerides** are derivatives of glycerol-3-phosphate (red), which contains two esterified fatty acid chains that constitute the hydrophobic "tail" and a polar "head" group.

Sphingolipids are derivatives of sphingosine (red), an amino alcohol with a long hydrocarbon chain.

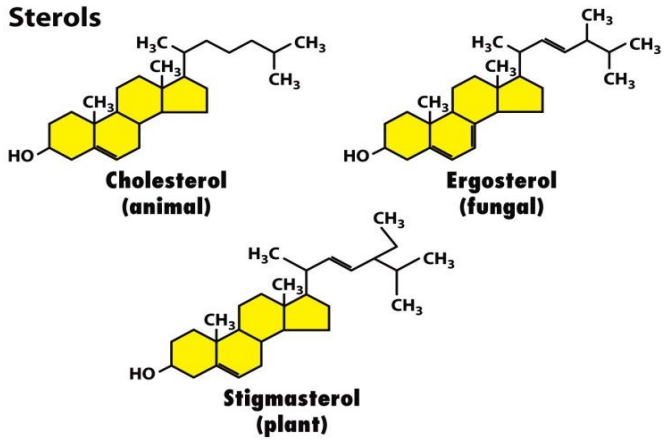


Figure 10-5c
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Basic structure of steroids is a four-ring hydrocarbon.

Formation and study of pure phospholipid bilayer

<u>Properties of bilayers:</u>

-**impermeable barrier** that prevents the diffusion of water-soluble

-The bilayer structure is maintained by hydrophobic and van der Waals interactions between the lipid chain -> **stability**

-All phospholipid bilayers can spontaneously form sealed closed compartment internal Aqueous Space

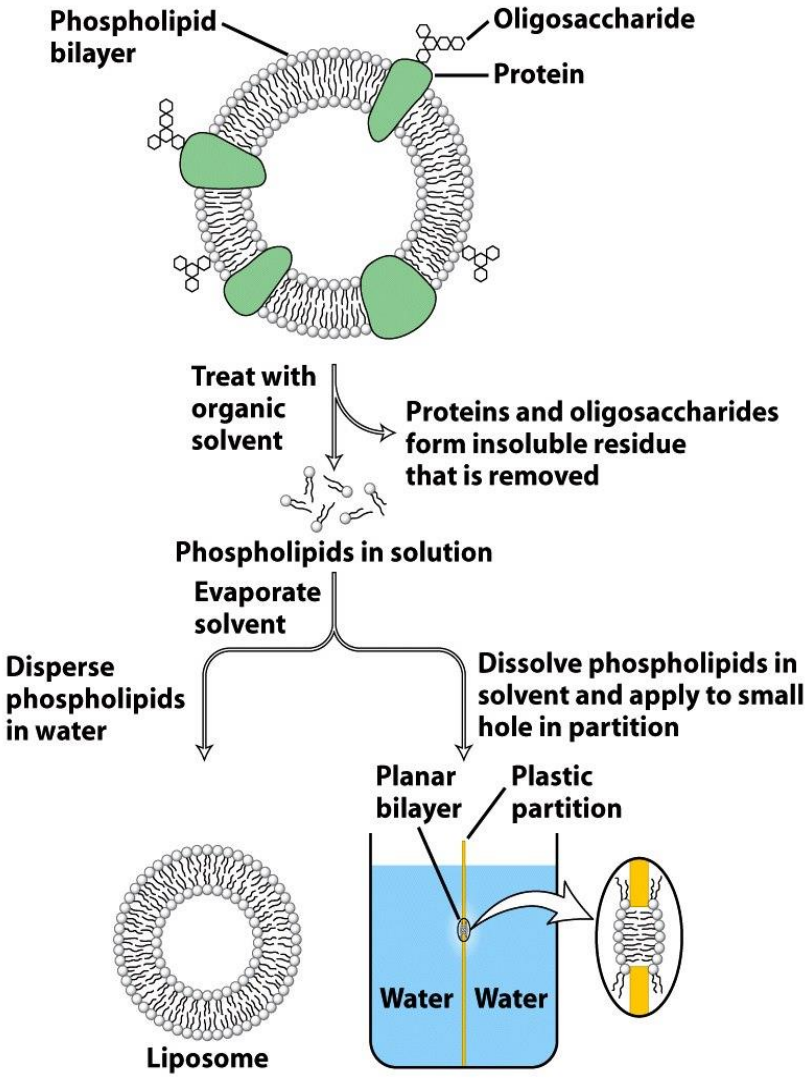


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Protein components

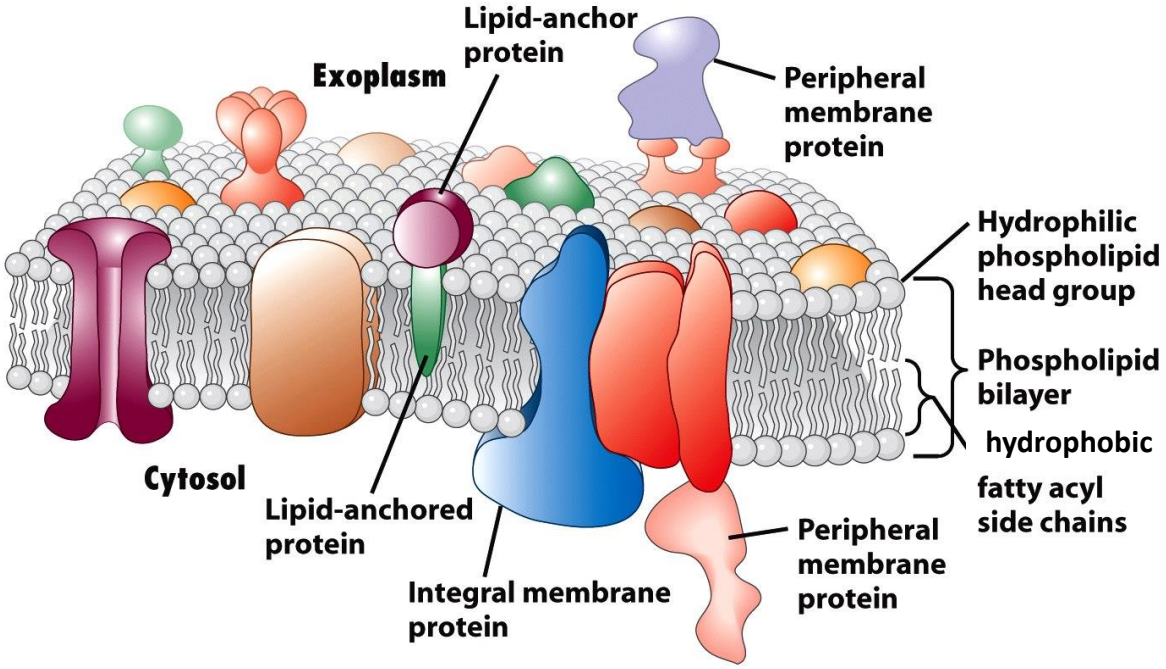
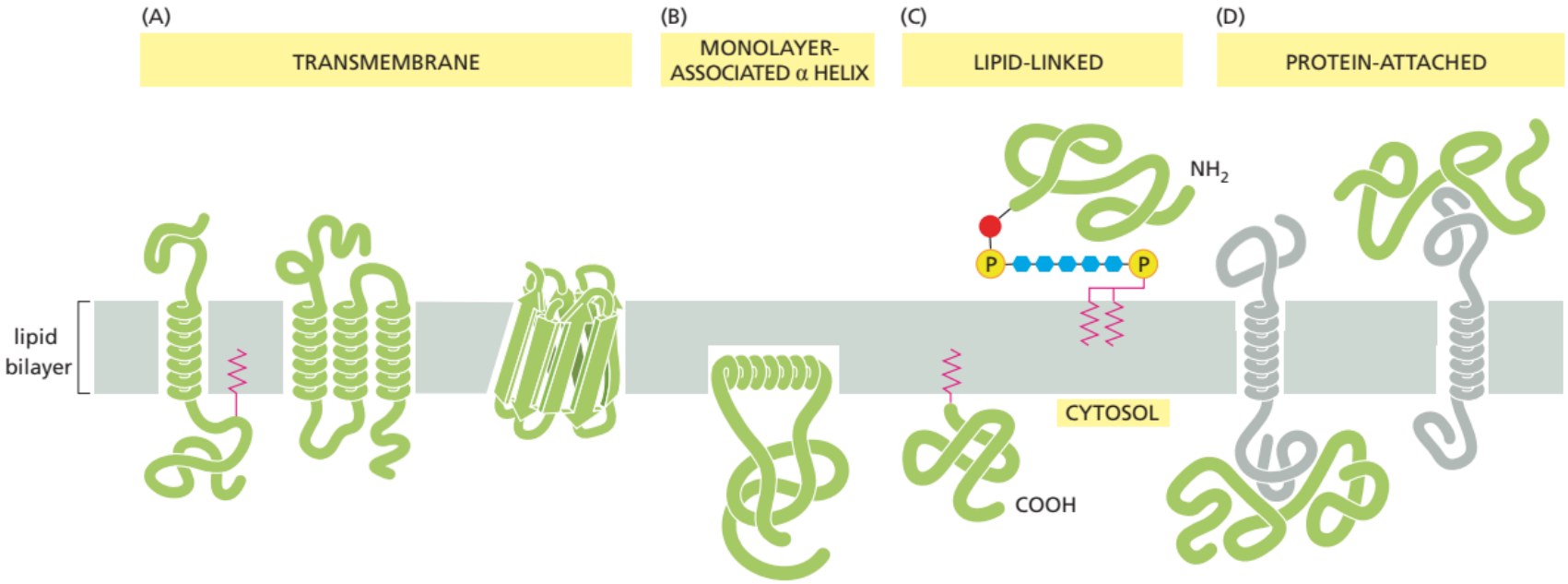


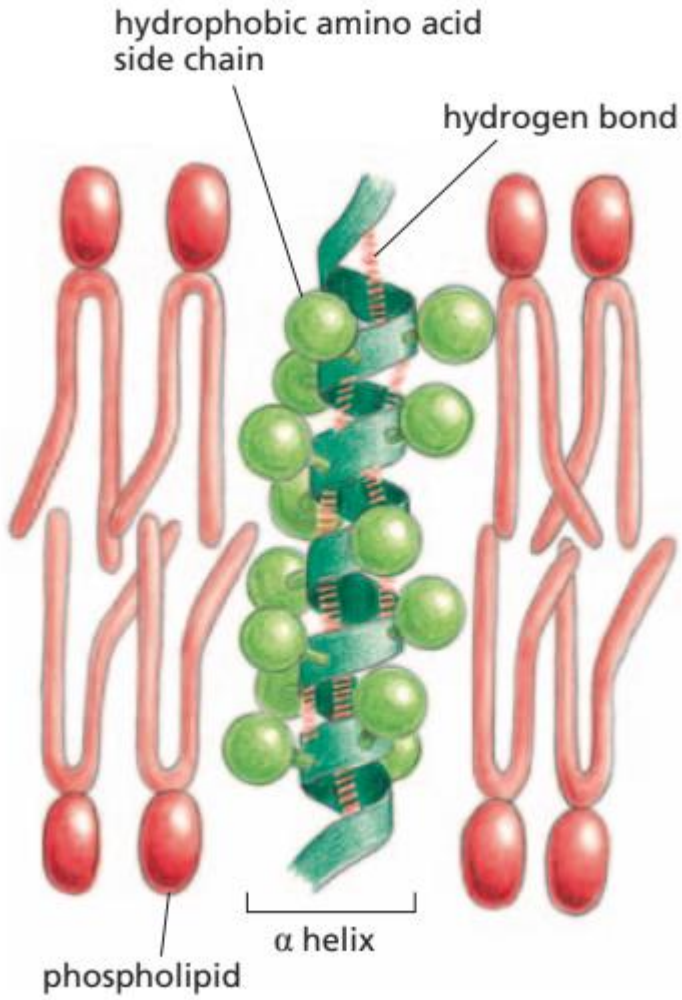
Figure 10-1
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Integral (protein transmembrane) proteins span the bilayer and often form dimers and higher-order oligomers.(>70% of the membrane proteins)

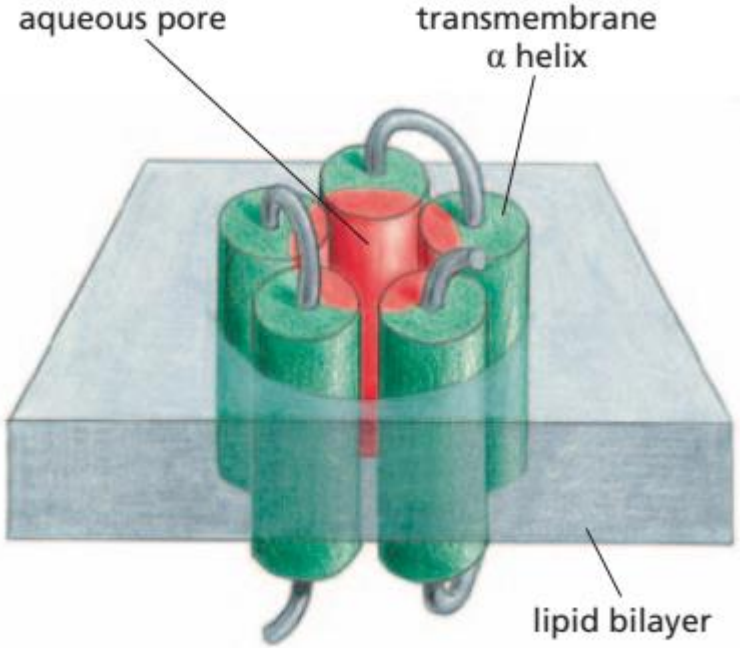
Peripheral proteins associate with the membrane primarily by specific noncovalent interactions with intergral proteins or membrane lipids.



Membrane proteins can associate with the lipid bilayer in different ways.
(A) Transmembrane proteins extend across the bilayer as a single α helix, as multiple α helices, or as a rolled-up β sheet (called a β barrel).
(B) Some membrane proteins are anchored to the cytosolic half of the lipid bilayer by an amphipathic α helix.
(C) Others are linked to either side of the bilayer solely by a covalently attached lipid molecule (red zigzag lines).
(D) Many proteins are attached to the membrane only by relatively weak, noncovalent interactions with other membrane proteins. all except (D) are integral membrane proteins.



A transmembrane polypeptide chain usually crosses the lipid bilayer as an α helix.



A transmembrane hydrophilic pore can be formed by multiple amphipathic α helices.

In this example, five such transmembrane α helices form a water-filled channel across the lipid bilayer. The hydrophobic amino acid side chains (green) on one side of each helix contact the hydrophobic lipid tails, while the hydrophilic side chains (red) on the opposite side of the helices form a water-filled pore

Human ABO blood group antigen

These antigens are oligosaccharide chains covalently attached to glycolipids or glycoproteins in the plasma membrane.

The terminal oligosaccharide sugars distinguish the three antigens.

The presence or absence of the glycosyl transferases that add (Gal) or (GalNAc) to O antigen determine a person's blood type

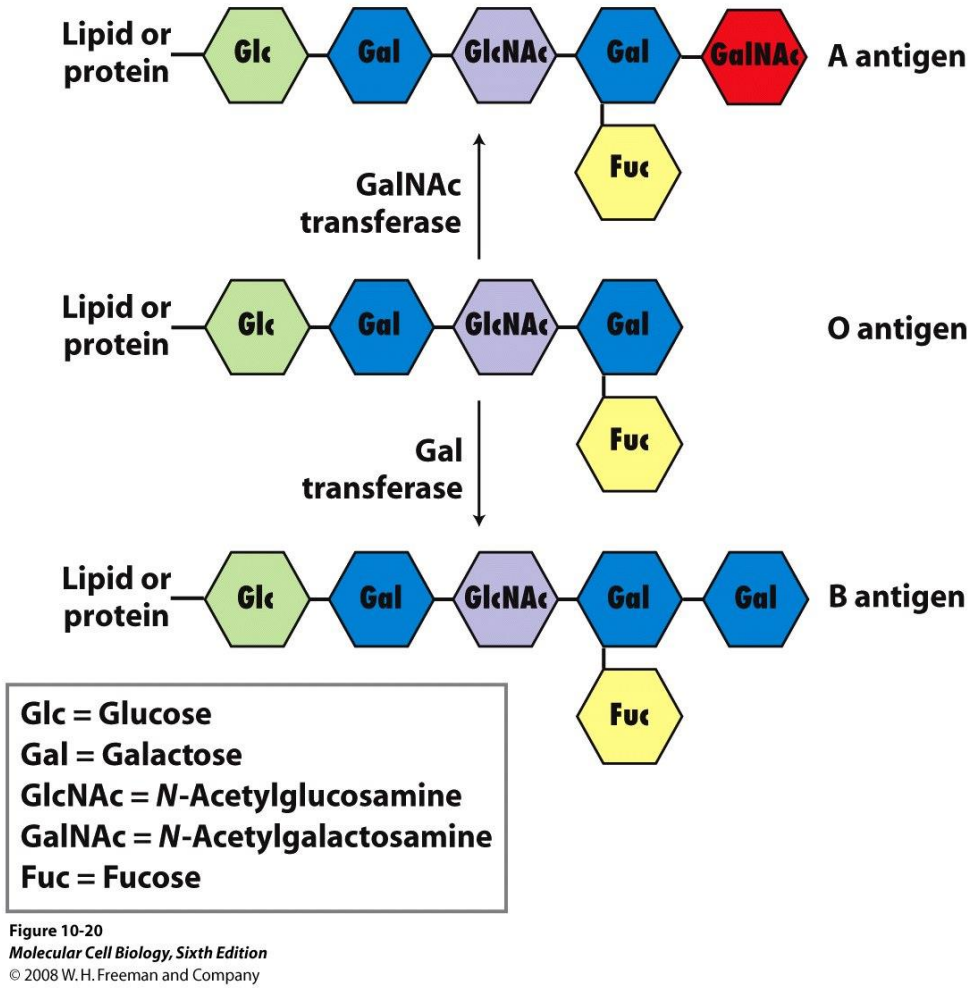


Figure 10-20
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TABLE 10-2 ABO Blood Groups

BLOOD GROUP	ANTIGENS ON RBCS*	SERUM ANTIBODIES	CAN RECEIVE BLOOD TYPES
A	A	Anti-B	A and O
B	B	Anti-A	B and O
AB	A and B	None	All
O	O	Anti-A and anti-B	O

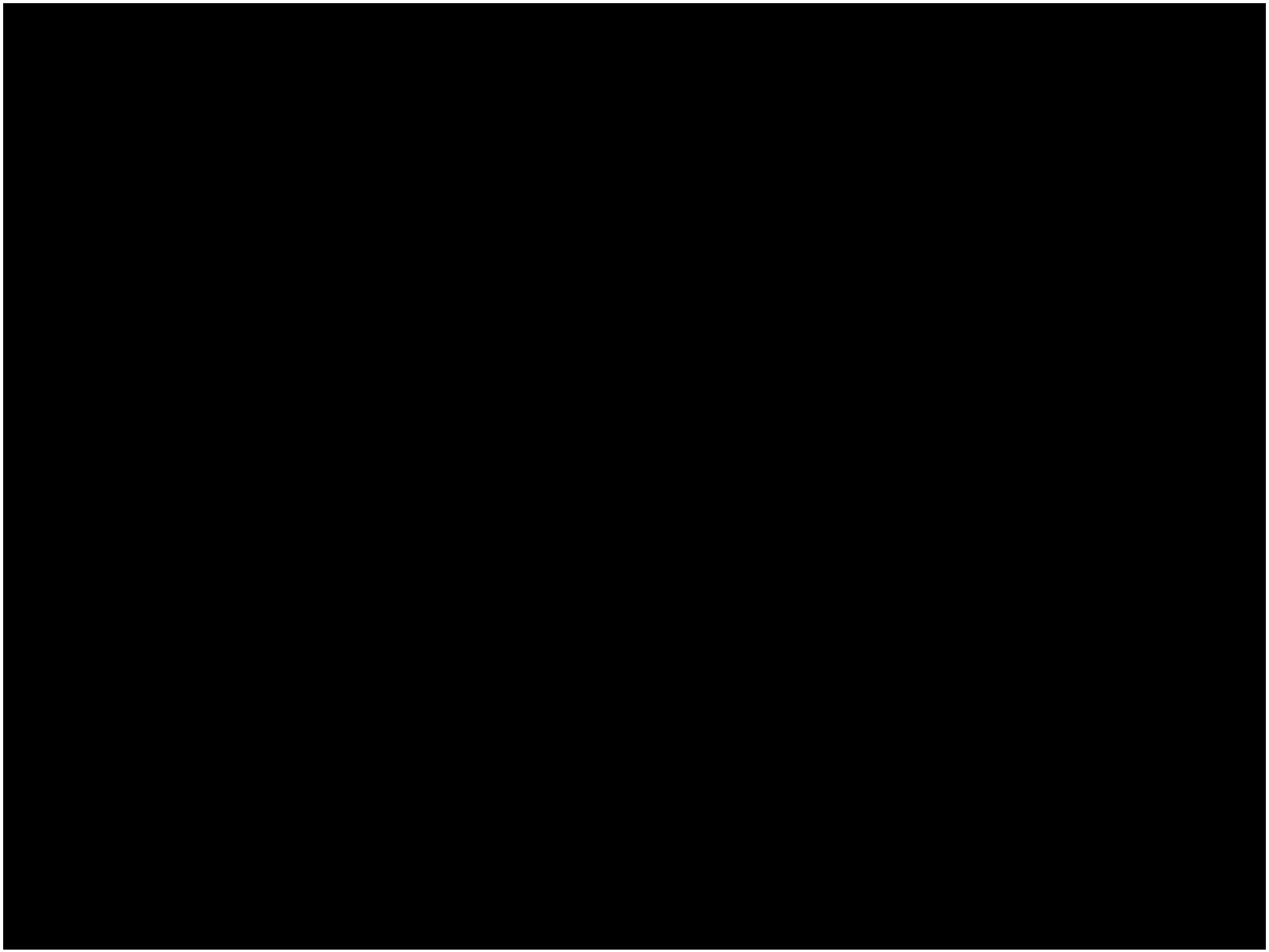
***See Figure 10-20 for antigen structures.**

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People whose erythrocytes lack the A antigen, the B antigen, or both on their surface normally have antibodies against the missing antigen(s)in their serum

Ex. if a type A or O person receives a transfusion of type B blood, antibodies against the B antigen will bind to the introduced red cells and trigger their destruction

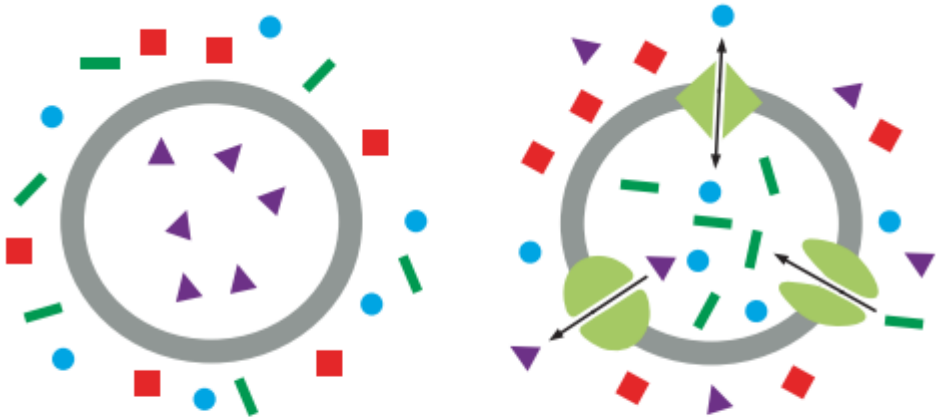
Cell membrane structure





Chapter 11 Opener
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Principles of Transmembrane Transport



Cell membranes contain specialized membrane transport proteins that facilitate the passage of selected small water-soluble molecules.

(a) protein-free, artificial lipid bilayers such as liposomes are impermeable to most water-soluble molecules.

(b) cell membranes, by contrast, contain transport proteins, each of which transfers a particular type of molecule. This selective transport can include the active pumping of specific molecules either out of (purple triangles) or into (green bars) the cell. The combined action of different transport proteins allows a specific set of solutes to build up inside a membrane enclosed compartment, such as the cytosol or an organelle

Overview of Membrane Transport

Relative permeability of a pure phospholipid bilayer to various molecules.

A bilayer is **permeable** to small hydrophobic molecules and small uncharged polar molecules, **slightly permeable** to water and urea, and essentially **impermeable** to ions and to large polar molecules

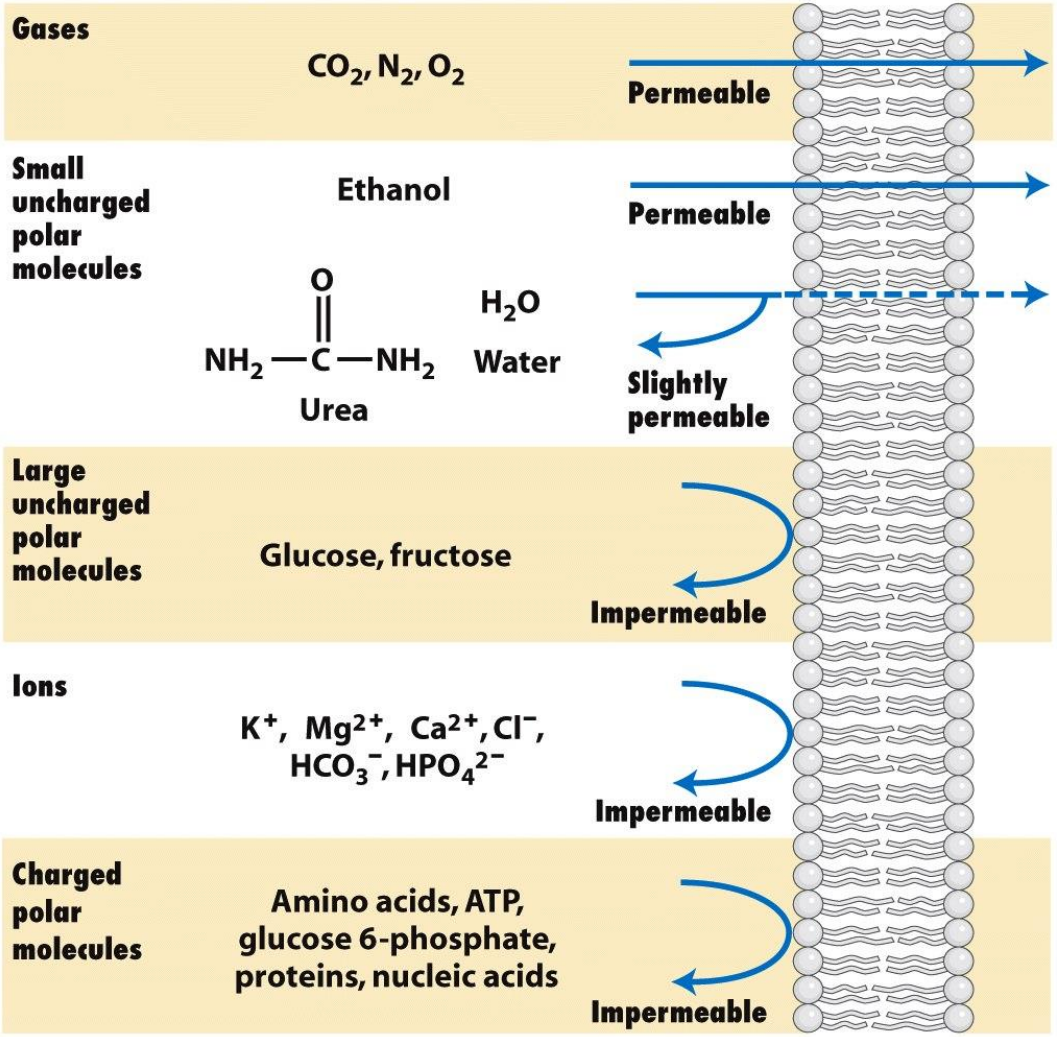
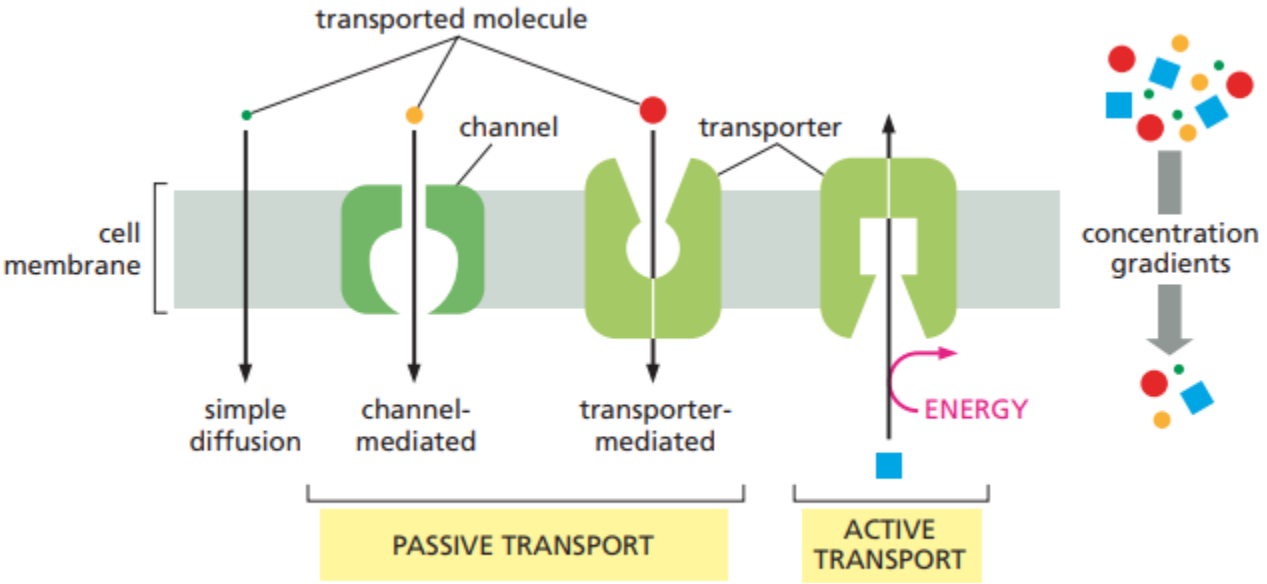


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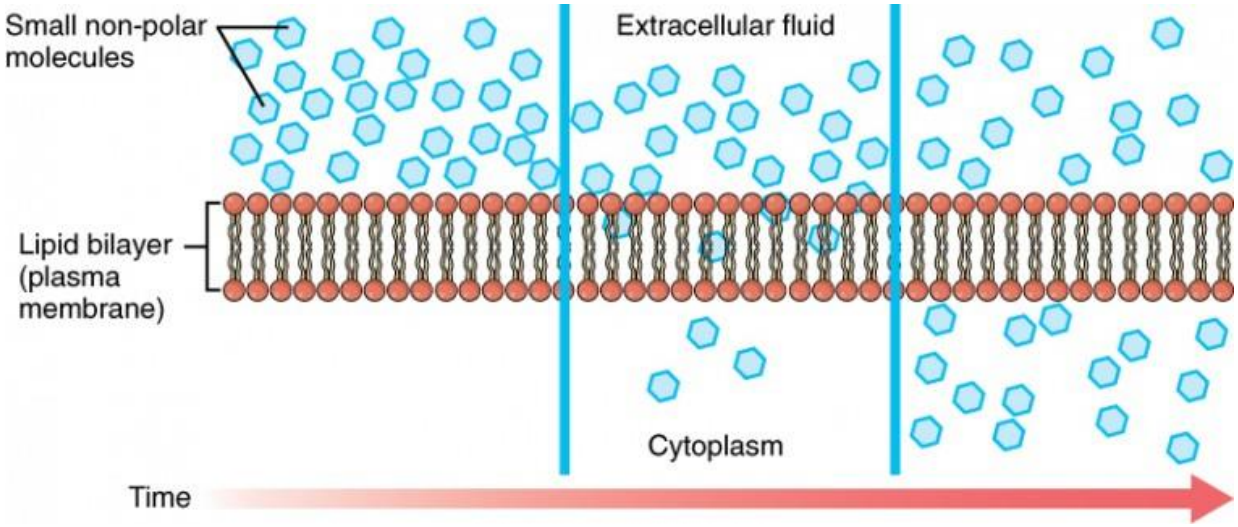
Solutes cross cell membranes by either passive or active transport.

1. Simple diffusion

Only small hydrophobic molecules cross membrane by **simple diffusion**

-Simple diffusion is the transport or movement of molecules from higher concentration to lower concentration **without expenditure of energy**.

-Without aid of transport proteins



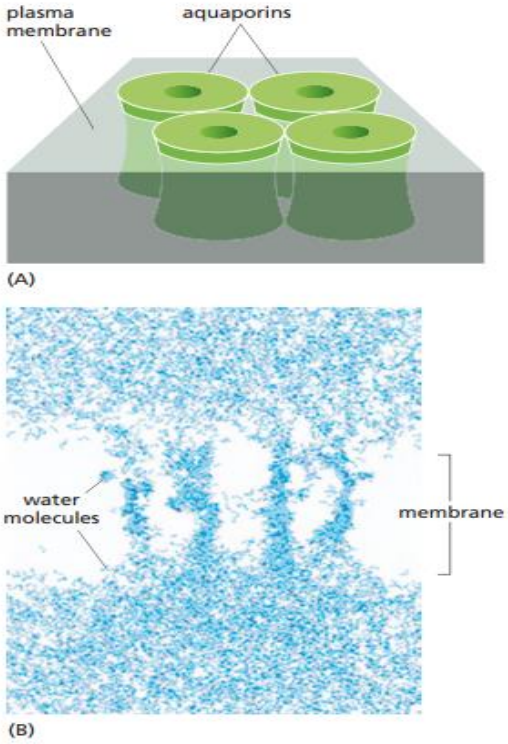
2. Facilitated diffusion

- Required transport protein

  - **Channel protein**: **Open channel** proteins: create a pore in the cell membrane and allow the charged molecules to pass through. **Gated channel** proteins are either closed or open and regulate the entry and exit of substances.
  - **Carrier Proteins**: change the confirmation of the molecules and release the molecules to the other side.

- Example:

  - Aquaporin:
  - Ion channels: allow the selective transport of ions and solutes across the plasma membrane
  - Glucose Transporter



Aquaporin



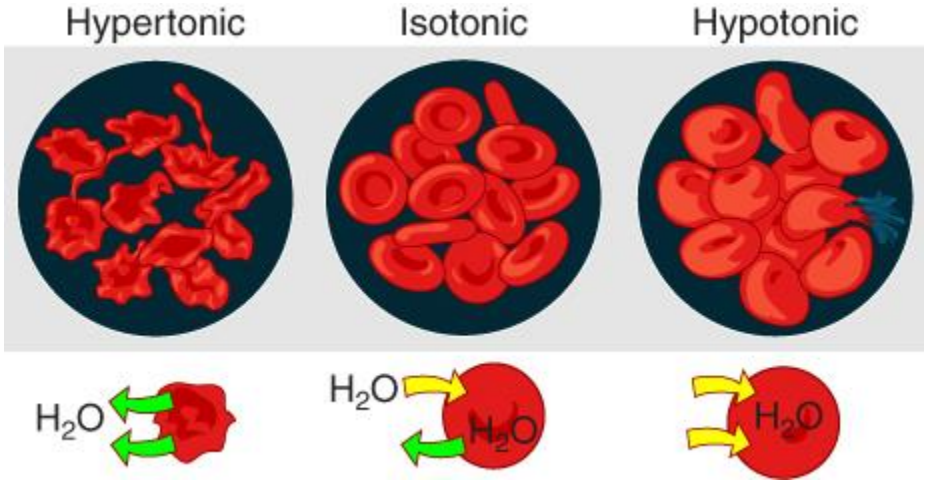
Osmosis

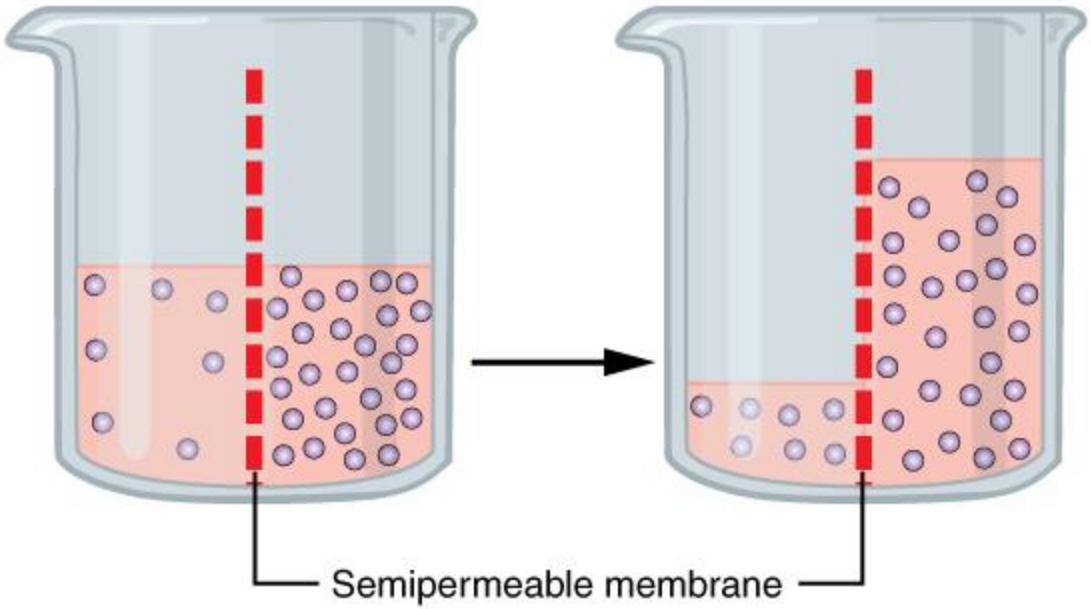
The movement of solvent (water) across membrane in response to the concentration gradient of solute is known as Osmosis

In **isotonic** (đẳng trương) solution, water move equally in both direction inside and outside of cell.

In **hypertonic** (ưu trương) solution, water moves out of the cell so that cell shrinks.

In **hypotonic** (nhược trương) solution, water moves inside the cell so that cell swells up.





Osmosis is the diffusion of water through a s**emipermeable membrane** down its concentration gradient

3. Active transport: energy required

Pumps utilize the energy released by ATP hydrolysis to power movement of specific ions (red circles) or small molecules against their electrochemical gradient/ **Active transport**

Channels permit movement of specific ions (or water) down their electrochemical gradient/ **Ion channels – passive transport**

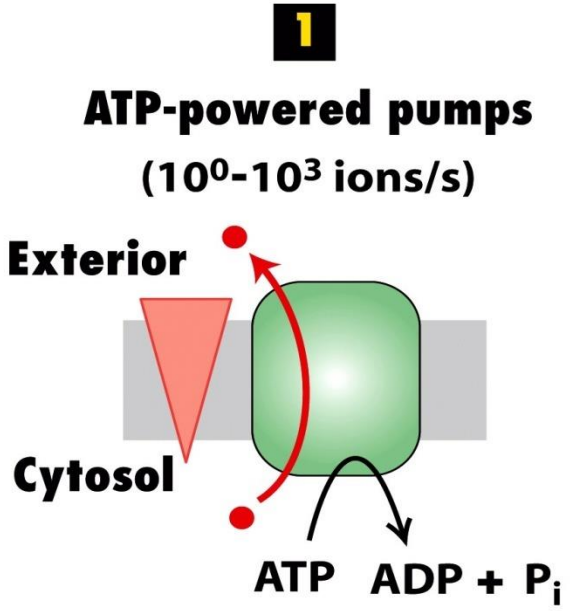


Figure 11-3 part 1

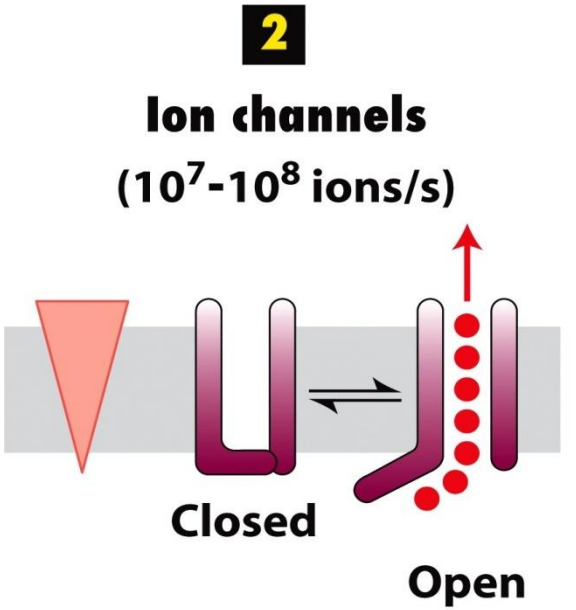


Figure 11-3 part 2
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Gradients are indicated by triangles with the tip pointing toward lower concentration, electrical potential, or both.

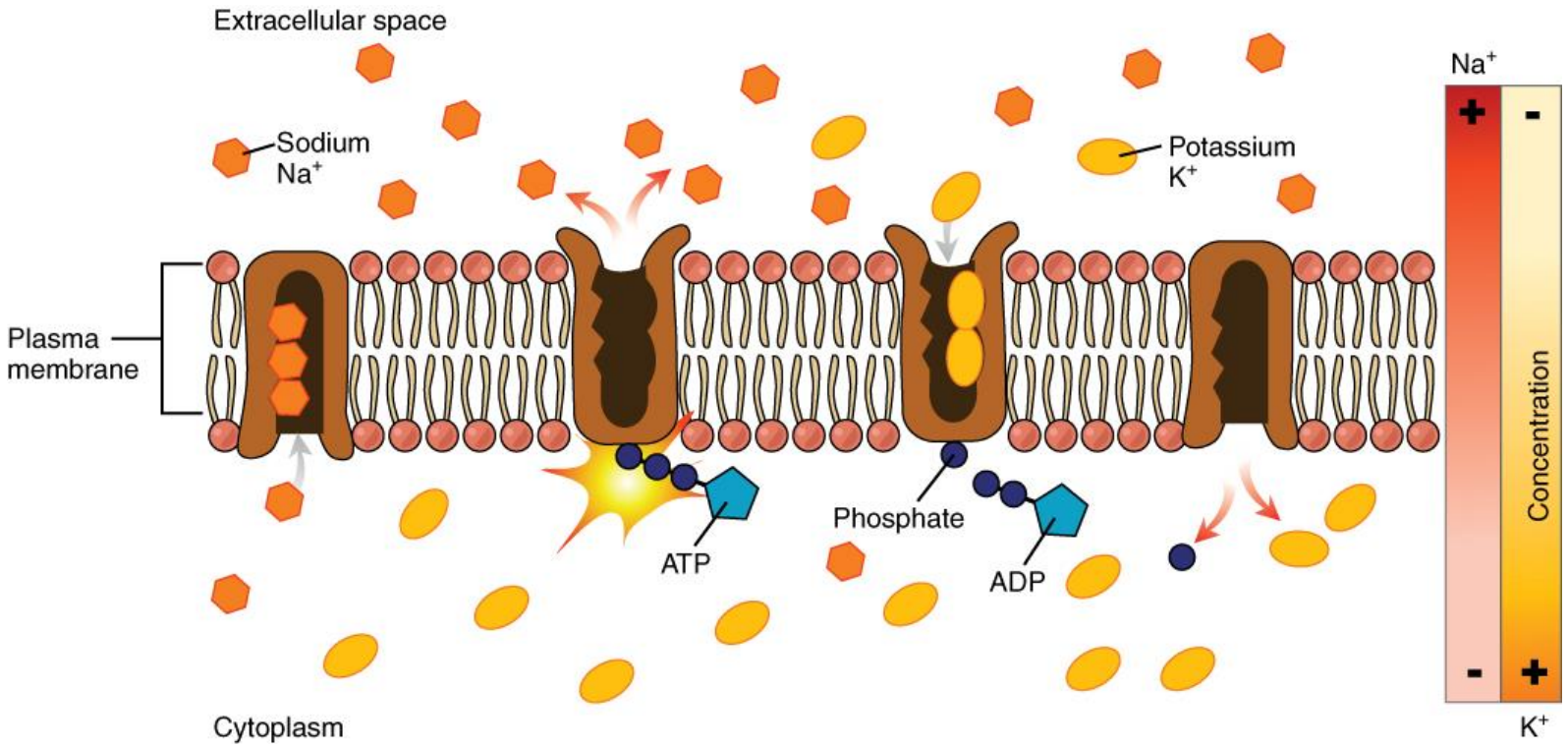
Differences in the concentration of inorganic ions across a cell membrane creates a **membrane potential**

TABLE 12–1 A COMPARISON OF ION CONCENTRATIONS INSIDE AND OUTSIDE A TYPICAL MAMMALIAN CELL

Component	Intracellular Concentration (mM)	Extracellular Concentration (mM)
Cations		
Na^+	5–15	145
K^+	140	5
Mg^{2+}	0.5*	1–2
Ca^{2+}	10^{-4}*	1–2
H^+	7×10^{-5} ($10^{-7.2}$ M or pH 7.2)	4×10^{-5} ($10^{-7.4}$ M or pH 7.4)
Anions**		
Cl^-	5–15	110

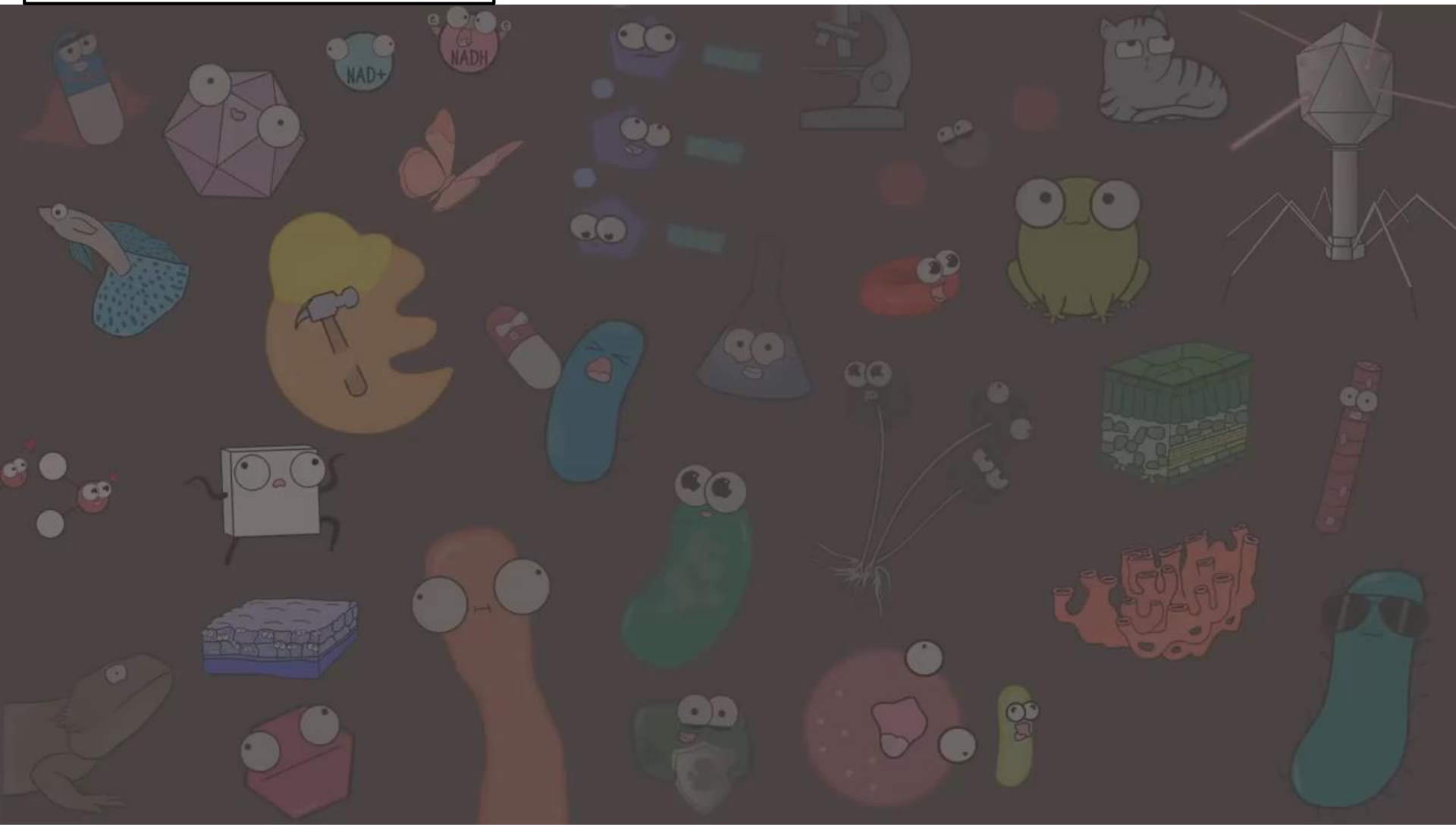
*The concentrations of Mg^{2+} and Ca^{2+} given are for the free ions. There is a total of about 20 mM Mg^{2+} and 1–2 mM Ca^{2+} in cells, but these ions are mostly bound to proteins and other organic molecules and, for Ca^{2+}, stored within various organelles.
**In addition to Cl^-, a cell contains many other anions not listed in this table. In fact, most cell constituents are negatively charged (HCO_3^-, PO_4^{3-}, proteins, nucleic acids, metabolites carrying phosphate and carboxyl groups, etc.).

Sodium-Potassium Pump



Powered by ATP, the pump moves sodium and potassium ions in opposite directions, each against its concentration gradient. In a single cycle of the pump, **three sodium ions** are extruded from and **two potassium ions** are imported into the cell.

Sodium – potassium pump



Transporters, which fall into three groups, facilitate the movement of specific small molecules or ion

Uniporters transport a single type of molecule down its concentration gradient **A**.

Co-transport proteins (**symporters**, **B**, and **antiporters, C**) catalyze the movement of one molecule against its concentration gradient (black circles), driven by the movement of one or more ions down the electrochemical gradient (red circle)

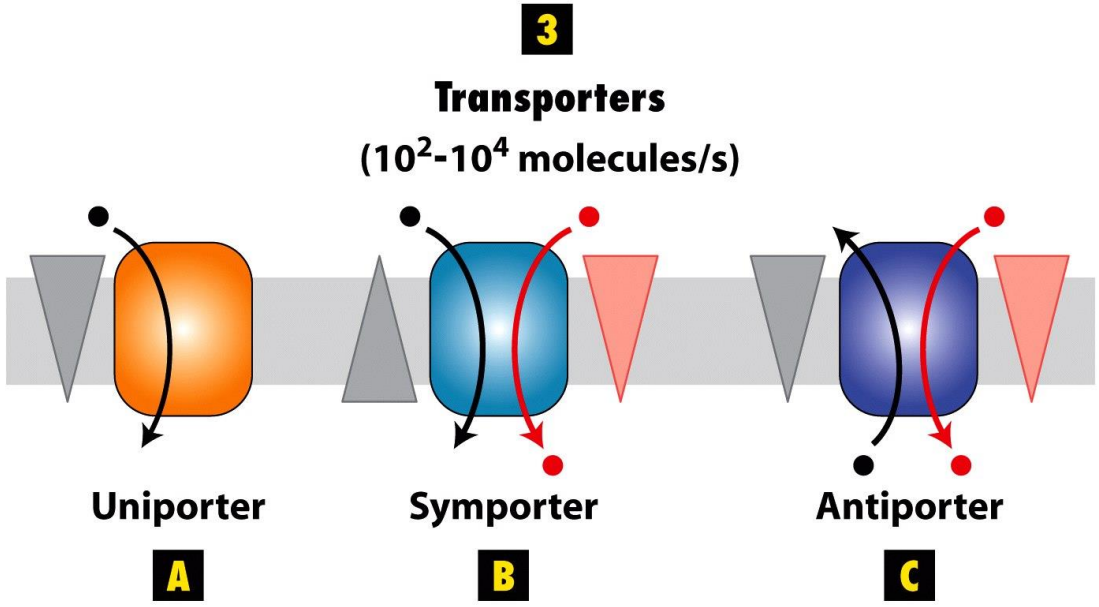
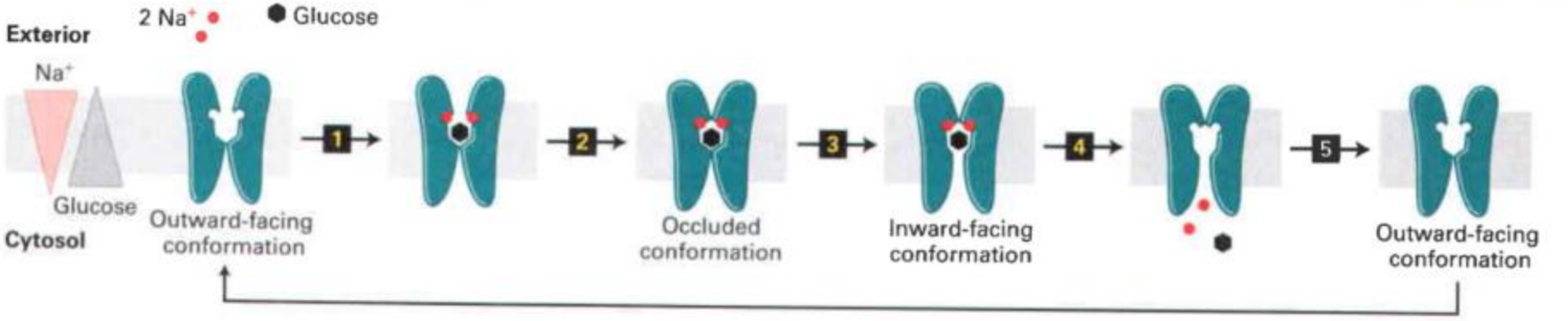


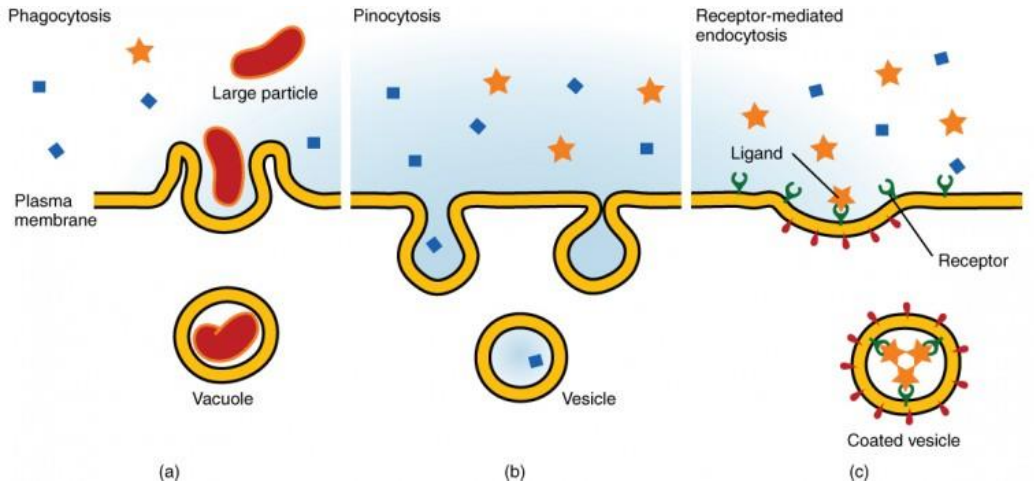
Figure 11-3 part 3
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Operational model for the two-Na+/one glucose symporter. Simultaneous binding of Na+ and glucose to the conformation without ward-facing binding sites (step 1) causes a conformational change in the protein such that the bound substrates are transiently occluded, unable to dissociate into either medium (step2) In step 3 the protein assumes a third conformation with inward-facing sites Dissociation of the bound Na+ and glucose into the cytosol (step 4) allows the protein to revert to its original outward-facing conformation (step5), ready to transport additional substrate

Endocytosis

Endocytosis is a form of **active transport** in which a cell envelopes extracellular materials using its cell membrane



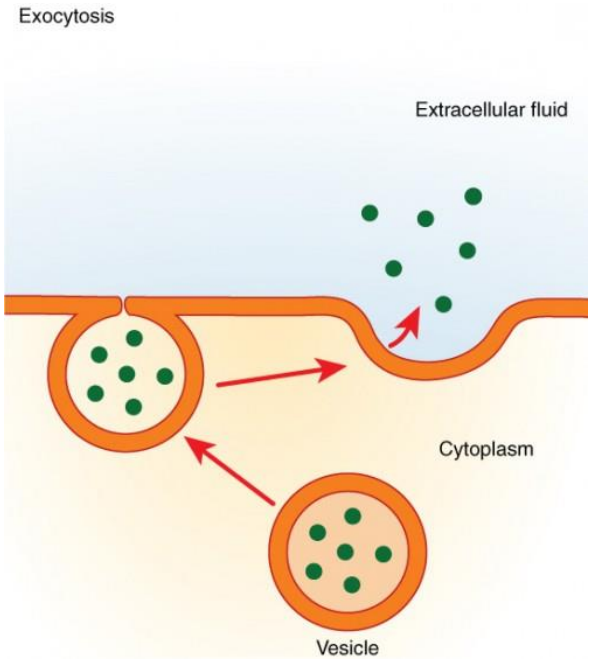
a) In **phagocytosis**, which is relatively nonselective, the cell takes in a large particle.

b) In **pinocytosis**, the cell takes in small particles in fluid

c) Receptor-mediated endocytosis is quite selective. When external receptors bind a specific ligand, the cell responds by endocytosing the ligand.

Exocytosis

The process of a cell exporting material using vesicular transport



Exocytosis is much like endocytosis in reverse. Material destined for export is packaged into a vesicle inside the cell. The membrane of the vesicle fuses with the cell membrane, and the contents are released into the extracellular space.

TABLE 11-1 Mechanisms for Transporting Ions and Small Molecules across Cell Membranes

PROPERTY	SIMPLE DIFFUSION	FACILITATED TRANSPORT	ACTIVE TRANSPORT	COTRANSPORT*
Requires specific protein	–	+	+	+
Solute transported against its gradient	–	–	+	+
Coupled to ATP hydrolysis	–	–	+	–
Driven by movement of a cotransported ion down its gradient	–	–	–	+
Examples of molecules transported	O_2, CO_2, steroid hormones, many drugs	Glucose and amino acids (uniporters); ions and water (channels)	Ions, small hydrophilic molecules, lipids (ATP-powered pumps)	Glucose and amino acids (symporters); various ions and sucrose (antiporters)

*Also called *secondary active transport*.

Table 11-1
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